

Mario Alan Cantú Cantero

Machine Learning Engineer / Data Scientist

github.com/Mario-16180 | linkedin.com/in/mariocantu21 | mario.alan.16180@gmail.com | +528130860870**Education****Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)**

Abu Dhabi, United Arab Emirates

Master of Science in Machine Learning (CGPA 3.85/4.0)

August 2022 – May 2024

- **Relevant coursework:** Mathematical Foundations of Machine Learning, Machine Learning, Artificial Intelligence, Optimization, Probabilistic and Statistical Inference, Advanced Machine Learning
- **Thesis:** A new curriculum learning approach for off-policy deep reinforcement learning algorithms [Repo](#)

Instituto Tecnológico de Monterrey (ITESM)

Monterrey, Mexico

B.Sc. in Mechatronics Engineering with a minor in Applied Robotics (CGPA 97.53/100)

August 2016 – December 2021

- **Study abroad:** Universidad de Cádiz, Spain (GPA 96/100)
- **Distinctions:** summa cum laude (Honorable Mention of Excellence) & Diploma of Excellence in Integral Formation
- **Relevant coursework:** Intelligent Systems, Computer Vision
- **Selected project to showcase:** Automated drug delivery mobile robot assistant in a simulated environment.

Experience**IDS Comercial / Walmart Mexico**

Monterrey, Mexico

Mid-Level Machine Learning Engineer

August 2024 – Ongoing

- Deployed an upsell recommender model on GCP, leveraging Apache Spark for distributed processing. Built end-to-end ML pipelines for training and inference, with less than an hour runtime for 5M records. Orchestrated weekly input updates using Vertex AI and Kubeflow, enabling the marketing team to access the model when given input with an API. Delivered the project within a 2-month timeline, adhering to MLOps best practices.
- Contributed to ML workflows across 4+ projects, resolving critical bottlenecks and completing specific technical tasks. For instance, reduced inference time for a cosine similarity-based model from 36 hours to 70 minutes by refactoring the calculation logic and improving resource utilization. Provided mentorship and tutorials to team members.
- Implemented model monitoring solutions by integrating lightweight tracking code into inference pipelines. Captured key performance metrics (e.g., drift and metadata) and visualized them in Looker Studio, enabling detection of model degradation and improving system reliability.
- Contributed to an internal MLOps utility library to standardize pipeline creation, database operations (GCS, BigQuery), and credential management. This library reduced development time and ensured consistency across 7+ ML projects.
- Automated report generation using Python-based RPA tools, streamlining the creation of .xlsx reports and saving weekly hours for the analytics team.

Scale AI

Monterrey, Mexico

Freelance

June 2024 – August 2024

- Evaluated the performance of side-by-side LLM pairs that can solve different task types, for example, Python, SQL, and C/C++ code generation.
- Documented justifications of which model is better between a pair of LLMs.
- Programmed the solution of different tasks that are presented to the LLMs in several programming languages for its future fine-tuning.

Abu Dhabi National Oil Company (ADNOC)

Abu Dhabi, United Arab Emirates

Machine Learning Engineer Intern

June 2023 – July 2023

- Created a Q&A system using a RAG (Retrieval-Augmented Generation) approach and GPT3.5 as an LLM to answer questions about oil drilling reports. Proposed the main solution and taught others the method. Also delegated the main responsibilities of the project, like data cleaning and API creation, overlooking the progress of each of my coworkers. About 70 % of the questions were successfully answered. Showcased with an API using FastAPI. Developed in less than a month. [Repo](#)
- Implemented a time-series forecasting model and an exploratory data analysis for the company's inflow of ethane from 2019 to 2022. Found that most of the missing values were during the COVID-19 pandemic, i.e. 2020. Also found that there

is a constant decrease of ethane flow over the years, probably due to lack of maintenance. The MSE of the LSTM model for forecasting was $2.4161e-04$.

Neoris

Monterrey, Mexico

Data Engineer

January 2022 – July 2022

- Migrated databases using SQL Server for FEMSA. Worked collaboratively with a team of 4 members using scrum methodology as project management. The historical data from the previous (at least) 10 years were successfully moved and documented.

Metalsa

Monterrey, Mexico

Data Scientist Intern

December 2020 – December 2021

- Deployed an organizational network analysis with visualizations and a parametric study of the informal structure within the organization to detect high-potential employees using NetworkX and Alteryx. Delivered a report showing insights about inner-company relations.
- Programmed a fuzzy logic system to estimate the years of preparation that every employee needs to climb to the next level in the organization using performance and potential data. Discovered potential high-level employees. Delivered a project showing the people that have yet to advance in seniority.
- Designed a machine learning model to predict attrition in the organization with significant insights as a result and ~ 0.80 accuracy on a test set.

Professional Certificates

[Fundamentals of Reinforcement Learning](#) – Coursera, University of Alberta January 2023

Professional Development in Data Science – BEDU June 2021

- Analyzed a vehicle accident database of Mexico City. Implemented an unsupervised model called DBScan algorithm to suggest attention centers for the positions with the highest occurrence of car accidents. [Repo](#)

[Machine Learning](#) – Coursera, Stanford University January 2021

Skills and languages

Languages: native Spanish, C1 English, basic French, basic German

Programming languages: Python, SQL, MATLAB, C/C++, R

Others/frameworks: LifeTimes, Docker, Kubeflow, PyTorch, Tensorflow, Numpy, Spark, Polars, OpenAI's LLMs, Matplotlib, Seaborn, Scikit-Learn, Scipy, Pandas, OpenCV, Arduino, WandB, GCP (BigQuery, Vertex), FastAPI, NetworkX, Alteryx